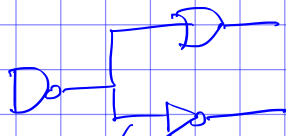


Interconnect

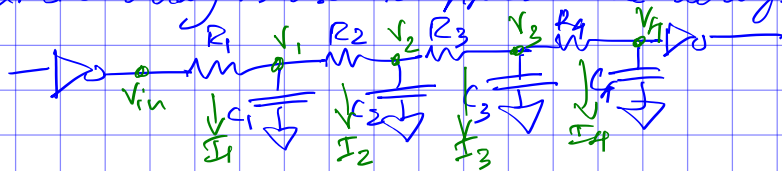


spans multiple layers



modelled as an RC ladder

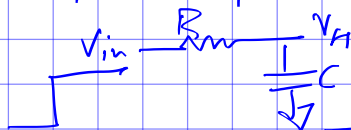
Elmore delay is used to approximate delay of interconnects.



$$I_i = C_i \frac{dV_i}{dt}$$

$$V_4 = V_{in} - R_1(I_1 + I_2 + I_3 + I_4) - R_2(I_2 + I_3 + I_4) - R_3(I_3 + I_4) - R_4 I_4$$

We want an effective time constant that approximates the transition at V_4 as though the RC ladder were a single RC



$\tau = RC$
time constant

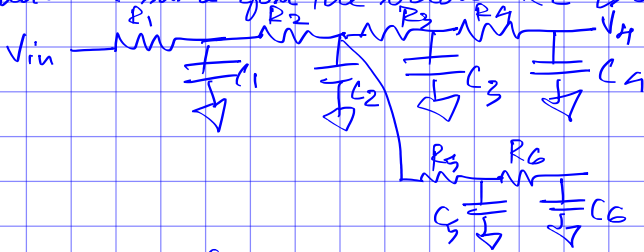
$$V_4 \approx V_{in} e^{-\frac{t}{\tau}}$$

$$\int_0^{\infty} e^{-\frac{t}{\tau}} dt = \int_0^{\infty} R_1 C_1 dV_1 + \int_0^{\infty} (R_1 + R_2) C_2 dV_2 + \int_0^{\infty} (R_1 + R_2 + R_3) C_3 dV_3 + \int_0^{\infty} (R_1 + R_2 + R_3 + R_4) C_4 dV_4$$

$$\tau = R_1 C_1 + (R_1 + R_2) C_2 + (R_1 + R_2 + R_3) C_3 + (R_1 + R_2 + R_3 + R_4) C_4$$

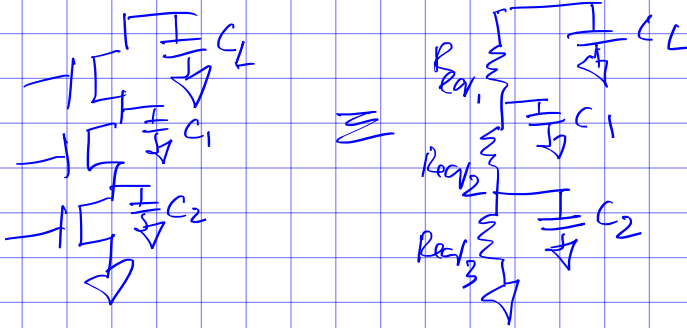
$$\tau = \sum_{k=1}^4 \left(\left(\sum_{i=1}^k R_i \right) C_k \right)$$

Repeat the same for the below RC tree.

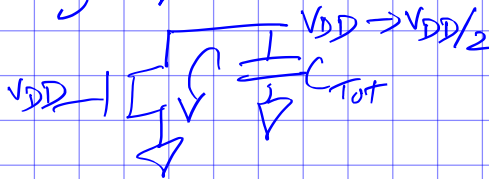


$$\tau = R_1 \sum_{i=1}^6 C_i + R_2 \sum_{i=2}^6 C_i + R_3 (C_3 + C_4) + R_4 C_4$$

We can analyze transistor stacks similar to the RC tree, if we can model the transistors as equivalent resistors.



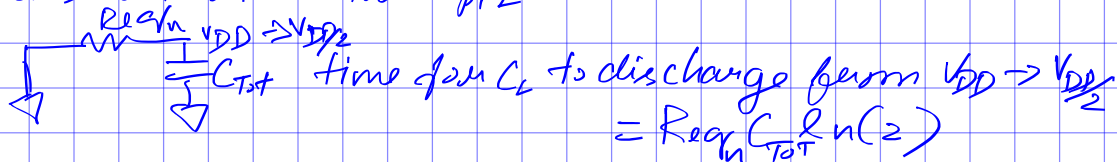
Finding equivalent resistance of transistor.



Let us consider the falling transition first.

$$t_{PHL} = \frac{C_{Tot} V_{DD}}{2 I_{DSATn}}$$

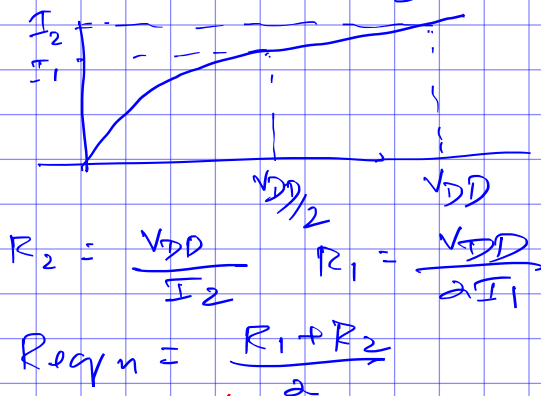
To find the equivalent resistance we match the time constant with the t_{PHL} .



$$R_{eq_n} = \frac{V_{DD}}{2 I_{DSAT_n} \ln(2)} \approx \frac{5}{7} \frac{V_{DD}}{I_{DSAT_n}}$$

Similarly $R_{eq_p} = \frac{5}{7} \frac{V_{DD}}{I_{DSAT_p}}$ } Ignoring channel length modulation.

To account for channel length modulation,



HW Find R_{eq_n} and R_{eq_p} for the unit inverter.

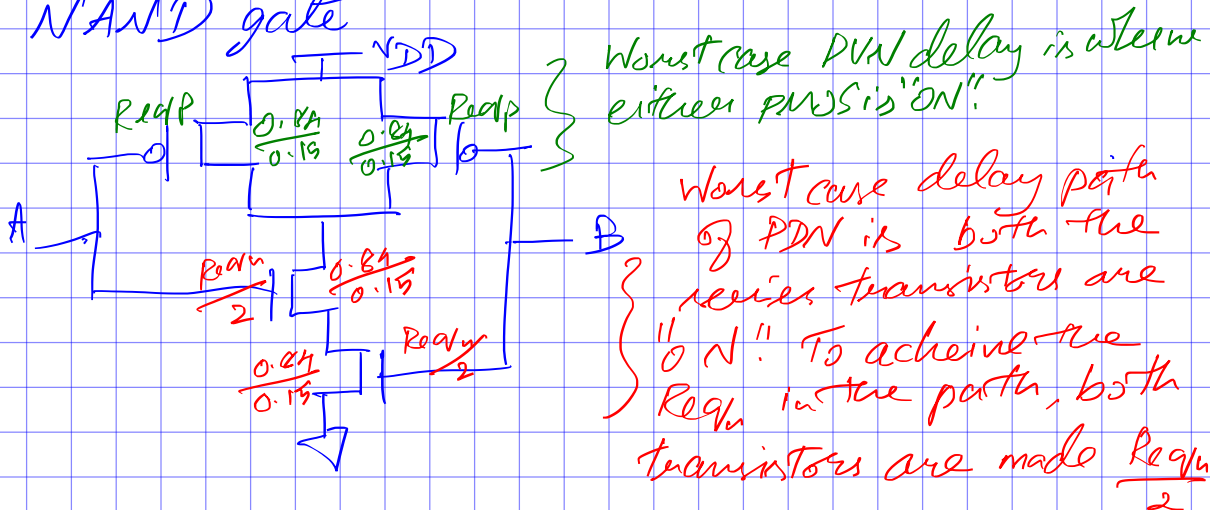
To scale up or down the resistance of the n/p mos the width is accordingly changed.
 e.g to get a $\frac{R_{eq_n}}{2}$, width of the nmos is scaled by a factor of 2.

Sizing of transistors in CMOS gates.

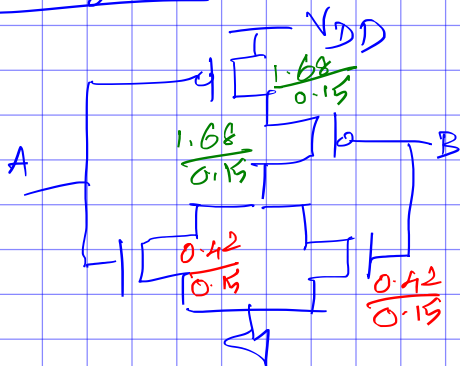
In all the following analysis we ignore the intermediate node caps.

To size the transistors, the resistance of the pull up and pull down networks are matched with that of an unit inverter for the worst case delay path.

NAND gate



NOR gate



Same argument holds good for NOR gate and is sized similarly.

HW find the sizes of transistors for $F = \overline{AB + C}$,
 $F = (A + B)(C + D)$, $F = \overline{AB + CD}$